AIMS
Airbus Material Specification

Carbon Fibre Reinforced Epoxy Prepreg Unidirectional Tape/180ºC - curing class High temperature

Standard modulus fibre
Structural material

Material Specification

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Contents

1 Scope

This Material Specification defines the requirements of the standard modulus carbon fibre reinforced unidirectional tape prepreg, curing 180ºC, high temperature for aerospace application.
The requirements specified in this Material Specification shall not be used as stress allowables.
Unless otherwise specified by the drawing, the purchase order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

This material is used for manufacturing primary airframe and aerospace structures (sandwich and monolithic)
The prepreg shall have an operating temperature range of –55ºC to 120ºC.
The material is to be delivered according to ABS 5347. The nominal cured ply thickness is 0.184 mm, related to a fibre volume fraction of 60 %.

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS 05-01-000 Part 1	Airbus Material Specification. General requirements. Carbon fibre reinforced epoxy prepreg. Technical Specification
AIMS 05-01-000 Part 7	Airbus Material Specification. Carbon fibre reinforced epoxy prepreg. Qualification program/ batch release testing. Unidirectional tape / 180ºC curing class. High temperature. Structural material. Technical Specification
AIMS05-01-000 Part 100	Carbon fiber reinforced epoxy prepreg Structural material Acceptance Testing and Storage Life Prolongation
ABS 5347	Airbus Standard. Carbon fibre reinforced epoxy prepreg. Unidirectional tape 180ºC – curing class. Standard modulus fibre. High temperature service. Structural material
AITM 1.0001	Airbus Test Method. Fibre reinforced plastics. Determination of mechanical degradation due to chemical paint strippers
AITM 1.0002	Airbus Test Method. Fibre reinforced plastics. Determination of in-plane shear properties
AITM 1.0003	Airbus Test Method. Fibre reinforced plastic materials. Determination of the glass transition temperatures
AITM 1.0005	Airbus Test Method. Carbon fibre reinforced plastics. Determination of interlaminar fracture toughness energy. Mode I
AITM 1.0006	Airbus Test Method. Carbon fibre reinforced plastics. Determination of interlaminar fracture toughness energy. Mode II
AITM 1.0007	Airbus Test Method. Fiber reinforced plastics. Determination of plain, open hole and filled hole tensile strength
AITM 1.0008	Airbus Test Method. Fiber reinforced plastics. Determination of plain, open hole and filled hole compression strength
AITM 1.0009	Airbus Test Method. Fibre reinforced plastics. Determination of bearing strength
AITM 1.0010	Airbus Test Method. Fibre reinforced plastics. Determination of compression strength after impact
AITM 3.0008	Airbus Test Method. Determination of the extent of cure by differential scanning calorimetry
ASTM E831	Standard test method for linear thermal expansion of solid materials by thermomecanical analysis
EN 2378	Aerospace series. Fibre reinforced plastics. Determination of water absorption by immersion
EN 2557	Aerospace series. Carbon fibre preimpregnates. Determination of mass per unit area
EN 2558	Aerospace series. Carbon fibre preimpregnates. Determination of the percentage of volatile matter.
EN 2559	Aerospace series. Carbon fibre preimpregnates. Determination of the percentage of

	resin and fibre content and the mass of fibre per unit area.
EN 2560	Aerospace series. Carbon fibre preimpregnates. Determinantion of the resin flow.
EN 2561	Aerospace series. Carbon fibre reinforced plastics unidirectional laminates. Tensile test paralell to the fibre direction
EN 2563	Aerospace series. Unidirectional laminates. Determination of the apparent interlaminar shear strength
EN 2823	Aerospace series. Unidirectional laminates carbon thermosetting resin. Tensile test perpendicular to the fibre direction
EN 2850	Aerospace series. Fibre reinforced plastics. Determination of the effect of exposure to humid atmosphere on physical and mechanical characteristics
EN 2597	Aerospace series. Carbon fibre thermosetting resin unidirectional laminates compression test parallel to the fibre direction
EN 3615	Aerospace series. Fibre Reinforced plastics. Determination of the conditions of exposure to humid atmosphere and moisture absorption
ISO 1183	Plastics. Mehtods for determining the density and relative density of non-cellular plastics
ISO 10119	Carbon fibres. Determination of density

4 Technical requirements

4.1 General

Specification values are minimum except where stated otherwise.

4.1.1 Material description

See Table 1

4.1.2 Worklife and storage conditions

See Table 2

4.1.3 Processing

The curing cycle shall be agreed between the manufacturer and Airbus at the onset of the material qualification programme and shall be used to produce all test specimens for qualification testing. The laminates for test purposes shall be cured in an autoclave. The curing cycle shall comply with the general requirements reported in Table 3.
The agreed curing cycle shall be specified in the relevant IPS and shall be used for the manufacturing of batch release test specimens.

4.1.4 Hazardous constituents

During manufacturing the prepreg shall not create any difficulties regarding health and safety or regarding the environment. The material shall be usable without any unusual specific protection procedures.
The relevant local health and safety/environmental protection laws shall be fulfilled.

4.2 Specific

4.2.1 Physical-chemical properties

Uncured and cured properties see Table 4.

4.2.2 Mechanical properties

See Table 5.

5 Designation

Refer to ABS 5347

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS 05-01-000 Part 1 and Part 7.

7 Incoming inspection

Materials qualified against this AIMS shall undergo “Technical Incoming Inspection” as per AIMS05-01-000 Part 100 prior to use.

Table 1: Material description

Characteristic	Requirements
Material type	UD carbon fibre tape / epoxy resin
Formulation	Fibre: Standard modulus carbon fibre Resin: Epoxy, toughened, high temperature
Form and method of production	Prepreg, cured in an autoclave at 180ºC
Technical Specification	AIMS 05-01-000 Parts 1 and 7
Resin content by weight, uncured	34% nominal
Service temperature range	-55ºC up to 120ºC

Table 2: Worklife and storage conditions

No.	Properties	Shop condition	Symbol	Unit	Requirements
1	Storage life at -18ºC	-	-	Months	> 12
2	Worklife 1) (tack life)	30ºC/75% R.H.	-	Days	> 10
		(25+2)ºC/(45+10)% R.H.	-	Days	> 10
1) The prepreg shall be capable of use after being exposed for an additional minimum of 5 days at the above mentioned shop condition, provided that the lay-up is fully bagged and ready for use.					

Table 3: Processing conditions

Cure type	Autoclave		
Cure conditions	Heating rate	ºC/min	0,2 – 5,0
	Cure temperature	ºC	175 – 190
	Cure time	min	120 – 240
	Cure pressure	MPa	0,35 – 1,1
		bar	3,5 – 11
	Cooling rate	ºC/min	0,2 – 3,5
	Vacuum (abs)	MPa	0,005 – 0,1
Note: The use of an intermediate temperature and low pressure dwell is permissible if agreed between Manufacturer and Airbus.			

Table 4: Physical properties

No.	Properties	Test methods	Unit	Requirements [1)]
Uncured				
1	Prepreg areal weight	EN 2557C	g/m^2	293 ± 20
2	Fibre areal weight	EN 2559C	g/m^2	196 ± 8
3	Resin density	ISO 1183 Method A	g/cm^3	See relevant IPS
4	Fibre density	ISO 10119	g/cm^3	1,78 ± 0,04
5	Volatile content	EN 2558	%	< 2
6	Resin content	EN 2559C	%	34 ± 2
7	Physic-chemical tests	AIMS 05-01-000 Part 1	-	See relevant IPS
8	Resin flow	EN 2560	%	See relevant IPS
9	Tack	To be agreed between manufacturer and Airbus	-	See relevant IPS
Cured				
10	Coefficient of thermal expansion	ASTM E 831	-	See relevant IPS
11	Water uptake	EN 3615, 70ºC/85% R.H./eq.	%	< 1,4
12	Water uptake	EN 2378, 70ºC/water/eq.	%	See relevant IPS
13	Extent of cure	AITM 3-0008	%	> 90
[1)] The stated requirements are average values.				

Table 5: Mechanical properties

No.	Properties	Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirements 1) X	Xmin
Material Code						**AIMS05-01-007**	
1	Glass transition temperature, Tg onset	AITM 1.0003	ºC	Dry	-	-	175
				Wet	-	-	145
2a	Tensile strength 0º	EN 2561 type B	MPa	Dry	-55	1800	1650
					RT	1800	1650
					120	1780	1630
				Wet	70	1670	1450
					120	1670	1450
				Water	120	1670	1450
2b	Tensile modulus 0º	EN 2561 type B	GPa	Dry	-55	135 ± 15	
					RT	135 ± 15	
					120	135 ± 15	
				Wet	70	135 ± 15	
					120	135 ± 15	
				Water	120	135 ± 15	
2c	Poisson's Ratio	EN 2561 type B	-	Dry	RT	0.27	
				Wet	70	0.27	
					120	0.27	
3a	Tensile strength 90º	EN 2597 type B	MPa	Dry	-55	55	50
					RT	60	50
					120	55	50
				Wet	70	30	25
					120	25	23
				Water	120	12	10
3b	Tensile modulus 90º	EN 2597 type B	GPa	Dry	-55	10 ± 1,7	
					RT	10 ± 1,7	
					120	8 ± 1,7	
				Wet	70	9,5 ± 1,7	
					120	7,7 ± 1,7	
				Water	120	7,7 ± 3,5	
4a	Compression strength 0º	EN 2850 type B	MPa	Dry	-55	1330	1210
					RT	1200	1135
					90	1100	1000
					120	1000	940
				Wet	70	1050	990
					120	800	750
				Water	120	750	700
4b	Compression modulus 0º	EN 2850 type B	GPa	Dry	-55	125 ± 15	
					RT	125 ± 15	
					90	125 ± 15	
					120	125 ± 15	
				Wet	70	125 ± 15	
					120	120 ± 20	
				Water	120	120 ± 20	

(Continued)

Table 5: Mechanical properties

No.	Properties	Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirements 1) X	Xmin
Material Code						**AIMS05-01-007**	
5a	Compression strength 90°	EN 2850 type B	MPa	Dry	RT	200	170
					120	150	130
				Wet	70	165	160
					120	110	100
5b	Compression modulus 90°	EN 2850 type B	GPa	Dry	RT	10 ± 1,5	
					120	8,25 ± 1,75	
				Wet	70	8,25 ± 1,75	
					120	8,00 ± 1,75	
6a	In plane shear strength	AITM 1.0002	Mpa	Dry	-55	90	85
					RT	90	85
					90	85	75
					120	80	70
				Wet	70	80	70
					90	75	65
					120	65	60
				Water	120	60	55
6b	In plane shear modulus	AITM 1.0002	GPa	Dry	-55	5 ± 1,5	
					RT	5 ± 1,5	
					90	4,8 ± 1,5	
					120	4,5 ± 1,5	
				Wet	70	4,5 ± 1,5	
					90	4,5 ± 1,5	
					120	3,0 ± 1,0	
				Water	120	2,5 ± 1,0	
7	Interlaminar shear strength 0°	EN 2563	MPa	Dry	-55	135	123
					RT	100	95
					90	70	65
					120	63	58
				Wet	70	68	63
					90	57	51
					120	53	45
				Water	120	45	41
8	Compression after impact 3)	AITM 1.0010	MPa	Dry	RT	160	
				Wet	120	140	
9	G1C	AITM 1.0005	J/m^2	Dry	RT	225	215
				Wet	RT	210	200
10	G2C	AITM 1.0006	J/m^2	Dry	RT	450	400
				Wet	RT	450	400

(Continued)

Table 5: Mechanical properties

No.	Properties	Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirements 1) X	Requirements 1) Xmin
Material Code						**AIMS05-01-007**	
11	**Resistance to aggressive fluids**						
11a	Interlaminar shear strength (reference)	EN 2563 + 4)	MPa	Dry	RT	80	72
				Wet	90	42	35
					120	42	35
11b	Interlaminar shear strength (skydrol, 70ºC/1000 h)	EN 2563 + 4)	MPa	Dry	90	6)	6)
11c	Interlaminar shear strength (skydrol/water, 50ºC/1000 h)	EN 2563 + 4)	MPa	Dry	90	6)	6)
11d	Interlaminar shear strength (fuel, RT/1000 h)	EN 2563 + 4)	MPa	Dry	90	6)	6)
11e	Interlaminar shear strength (MEK, RT/1 h)	EN 2563 + 4)	MPa	Dry	RT	6)	6)
12	**Resistance to paint strippers**						
12a	In plane shear strength (Phenolic paint stripper)	AITM 1.0001	MPa	Dry	RT	6)	6)
12b	In plane shear strength (Non phenolic paint stripper)	AITM 1.0001 + 5)	MPa	Dry	RT	6)	6)
13a	Plain tensile strength (unnotched)	AITM 1.0007 A Quasi-isotropic lay-up (25/50/25)	MPa	Dry	RT	575	500
					120	575	500
				Wet	120	575	500
13b	Open hole tensile strength (notched) 7)	AITM 1.0007 B Quasi-isotropic lay-up (25/50/25)	MPa	Dry	-55	285	260
					RT	285	260
				Wet	70	285	260
					120	285	260
13c	Plain tensile strength (unnotched)	AITM 1.0007 A Directed lay-up (40/40/20)	MPa	Dry	RT	680	600
					120	680	600
				Wet	120	680	600
13d	Open hole tensile strength (notched) 7)	AITM 1.0007 B Directed lay-up (40/40/20)	MPa	Dry	-55	380	350
					RT	380	350
				Wet	70	380	350
					90	380	350
					120	380	350
				Wet	70	250	225
					120	220	200
14a	Plain compression strength (unnotched)	AITM 1.0008 A Quasi-isotropic lay-up (25/50/25)	MPa	Dry	RT	625	580
					120	525	475
				Wet	120	440	400
14b	Open hole compression strength (notched) 7)	AITM 1.0008 B Quasi-isotropic lay-up (25/50/25)	MPa	Dry	RT	285	260
				Wet	70	250	225
					120	220	200
14c	Plain compression strength (unnotched)	AITM 1.0008 A Directed lay-up (40/40/20))	MPa	Dry	RT	700	625
					120	550	500
				Wet	120	500	460
14d	Open hole tensile strength (notched) 7)	AITM 1.0008 B Directed lay-up (40/40/20)	MPa	Dry	-55	375	340
					RT	350	315
				Wet	70	330	300
					120	270	245

(Continued)

Table 5: Mechanical properties (Concluded)

No.	Properties		Test method	Unit	Conditioning [2]	Test temp. [°C]	Requirements [1] X	Xmin
Material Code							AIMS05-01-007	
15a	Filled hole tensile strength [7]		AITM 1.0007 D Quasi-isotropic lay-up (25/50/25)	MPa	Dry	RT	285	260
					Wet	120	285	260
15b	Filled hole tensile strength [7]		AITM 1.0007 D Directed lay-up (40/40/20)	MPa	Dry	RT	350	315
						120	350	315
					Wet	70	350	315
						120	350	315
16a	Filled hole compression strength [7]		AITM 1.0008 D Quasi-isotropic lay-up (25/50/25)	MPa	Dry	RT	365	330
					Wet	70	265	240
						120	240	215
16b	Filled hole compression strength [7]		AITM 1.0008 D Directed lay-up (40/40/20)	MPa	Dry	RT	400	360
						120	315	285
					Wet	70	275	250
						120	250	225
17a	Bearing strength [7]	2% elong	AITM 1.0009 A Quasi-isotropic lay-up (25/50/25)	MPa	Dry	RT	800	700
						120	700	625
					Wet	70	550	500
						120	500	450
		Ult.			Dry	RT	900	800
						120	800	700
					Wet	70	650	600
						120	550	500
17b	Bearing strength [7]	2% elong	AITM 1.0009 A Directed lay-up (40/40/20)	MPa	Dry	-55	775	675
						RT	775	675
						120	650	600
					Wet	70	500	450
						120	440	400
		Ult.			Dry	-55	875	775
						RT	875	775
						120	750	675
					Wet	70	575	525
						120	460	420

1) Where applicable all values calculated with a theoretical thickness per ply of 0.184 mm with exception at ILSS, IPSS, IPSM, Tensile Strength 90º, Compression Strength 90º and CAI.
The stated requirements are average values: • X means minimun average value.
• Xmin means minimum individual value.

2) Conditioning "wet" is 70ºC/85% Relative Humidity according to EN2823 and "water" is 70ºC/water according to EN2378.

3) Residual Strength at 0.3 mm BVID indentation after relaxation.

4) AIMS 05-01-000 Part 1 para. 4.2.3.3

5) AIMS 05-01-000 Part 1 para. 4.2.3.4

6) These values will not have a decreasing higher than the corresponding one in comparison with the reference value at the equivalent temperature.

7) Requirements using gross section: (25/50/25)……….(+/0/-/90)3S
(40/40/20)……….(0/+/-/0/90)2S

Hole: 6,339-6,384 mm
Specimens width: 32 mm (Tensile and Compression), 45 mm (Bearing).
Filled hole specimens, fastener type:
Tensile and Compression: NAS 1154, ASNA2001 or equivalent, countersunk hole.
Bearing: NAS 6604-XX or an equivalent in terms of strength, material and surface finish.
Edge distance for bearing specimens 25 mm, hexagonal head steel alloy bolt finger tight.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 06/04		New Standard
2 06/05	Table 4	Change the tolerance of fibre areal weight from 196 ± 5 g/m^2 to 196 ± 8 g/m^2
3 07/13	Chapter 3 Chapter 7	AIMS05-01-000 part 100 added Chapter 7 defined in line with Incoming Inspection (Part 100) implementation